

PathFinder — A Personalised Learning Recommendation System

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The Problem in One Line

Our platform has thousands of courses, but learners don't know which one to pick — so they choose badly, get bored or stuck, and quit.

Our idea: instead of making learners *search* the catalogue, the platform should *guide* them — suggesting the right course, at the right difficulty, in the right order for their career goal. And to keep them going, they earn points that unlock real rewards.

Our goals:

Goal	Target
More engagement	+25%
More course completions	+30%
Happier users	+20%

How we'll know it worked: we launch it to only half the users first, then compare the two groups after 8 weeks.

Step 1 — User Stories

A) Learners

"Ananya, 26. Works in marketing, wants to become a data analyst. Studies 5 hours a week, mostly on her phone at night. She has started 4 courses and finished only 1."

L1. As a **learner**, I want to tell the app my career goal and how much time I have each week, so that I can get suggestions that fit my life instead of random popular courses.

L2. As a **learner**, I want the app to check my current skill level, so that I can avoid courses that are too easy (boring) or too hard (I give up).

L3. As a **learner**, I want a personalised home page that changes as I learn, so that I can find my next course in seconds.

L4. As a **learner**, I want my courses arranged as a step-by-step path with a progress bar, so that I can see how close I am to my goal.

L5. As a **learner**, I want to earn points and rewards for finishing modules, so that I can stay motivated and get something useful for my career.

B) Instructors (Course Creators)

"Rahul, 38. Has 6 courses on the platform. Wants more students and fair treatment for new courses."

- I1.** As an **instructor**, I want to tag my course with the skills it teaches and its difficulty level, so that it reaches the right students.
- I2.** As an **instructor**, I want to see how many people were shown my course, clicked it and finished it, so that I can fix the parts where students drop off.
- I3.** As an **instructor**, I want my new course to get a fair trial run of traffic, so that I can compete on quality and not lose just because I'm new.
- I4.** As an **instructor**, I want to see what skills students want but we don't teach yet, so that I can build courses people actually need.

C) Platform Admins

"Meera, 34. Head of Learning Operations. Responsible for platform numbers, quality and rules."

- A1.** As an **admin**, I want to change how courses are ranked using a settings page, so that I can adjust strategy without waiting for developers.
- A2.** As an **admin**, I want to A/B test changes before full launch, so that I can prove a change actually helped before rolling it out.
- A3.** As an **admin**, I want to check that we're not recommending the same few courses to everyone, so that new and smaller instructors still get seen.
- A4.** As an **admin**, I want to manage user privacy consent and data deletion, so that we follow GDPR and India's DPDP law.
- A5.** As an **admin**, I want to spot people cheating for points and set a budget limit on rewards, so that the rewards system doesn't get abused or become too expensive.

Step 2 — Functional Requirements

These are the 10 things the system must be able to do.

#	User Type	Feature	What It Does	Priority
F1	Learner	Sign-up questions + skill test	Asks the learner's goal, interests and free time (under 90 seconds), then a short quiz to find their level. Can be skipped.	High
F2	Learner	Personalised home feed	Shows the top 10 courses for that learner, in groups like <i>Continue Learning</i> and <i>Next Step for You</i> . Updates as they learn.	High
F3	Learner	Career path builder	Arranges 3-8 courses in the correct order (basics before advanced) with a progress bar and finish date.	High
F4	Learner	"Why this course?" + feedback buttons	Shows a simple reason on every card, plus buttons like <i>Not interested</i> and <i>I already know this</i> that improve future suggestions.	Medium
F5	Learner	Reminders and nudges	Sends a reminder when a learner goes quiet or is about to lose their streak, at the time of day they usually study.	High
F6	Learner	Points, streaks and rewards	Gives points for passing quizzes and finishing courses, tracks a weekly streak, and unlocks Bronze / Silver / Gold rewards. (<i>Details in Step 2A</i>)	Medium
F7	Instructor	Course tagging tool	Lets instructors tag skills, difficulty and prerequisites, with AI-suggested tags and a score showing how complete the tagging is.	High
F8	Instructor	Course performance dashboard	Shows views → clicks → enrolments → completions, and where students drop off. Also holds the mentor review queue.	Medium
F9	Admin	Settings + A/B testing panel	Lets admins change ranking rules, promote or hide courses (with an expiry date), and run experiments safely.	High
F10	Admin	Privacy, fairness and anti-cheating controls	Manages consent and data deletion, checks that recommendations stay varied, and detects point farming.	High

Why these priorities? F1, F2, F3 and F5 are the core — without them there is no product. F7, F9 and F10 are also High but for a different reason: bad course tags would break the suggestions, and without privacy controls we legally cannot launch. F4, F6 and F8 are Medium — very useful, but the system still works without them.

Step 2A — How the Rewards System Works (F6)

How you earn points

What you do	Points
Pass a module quiz	10
Submit a project	50
Finish a full course	100
Finish a full career path	300
Just watching a video	0

Why zero for watching videos? Because anyone can leave a video playing and walk away. We only give points for things you have to actually *prove* — like passing a quiz. Otherwise people would collect points without learning anything.

Streaks: you get a bonus if you hit your weekly study target several weeks in a row. We use a **weekly** streak, not daily, because our learners have jobs. You also get 2 "free passes" every 3 months, so one bad week doesn't destroy months of progress and make you quit.

The three reward levels

Level	Points needed	What you get
 Bronze	0-499	Profile badge + course certificate
 Silver	500+	Free 3-month subscription to an AI tool of your choice (ChatGPT Plus, Tableau, GitHub Copilot, etc.)
 Gold	2,000+ and top 5% of learners this quarter	A top industry mentor reviews your project and writes you a reference you can put on your resume + a 1-on-1 career call

Two important rules:

1. **No cash rewards.** If we gave cash, people would cheat to get it, and it would eat directly into our profit. Free tools cost us almost nothing because the tool companies want to reach learners anyway — and a tool actually *helps* you finish the next course.
2. **Points don't buy the reference.** Points only get you *eligibility*. A real mentor still reads your project and can say no. If everyone got a reference automatically, employers would stop believing it and the reward would be worthless.

Step 3 — Non-Functional Requirements

These are about *how well* the system works, not *what* it does.

Performance (speed)

- **The home page must load in under 0.3 seconds.** If the page is slow, people leave — and then our "more engagement" goal fails, no matter how good the suggestions are.

- **Points must appear within 5 seconds** of finishing a quiz. A reward you don't see immediately doesn't feel like a reward.

Scalability (handling growth)

- Must handle **10 million learners** and **5,000 people opening the app per second** at peak times (evenings and New Year).
- Must handle **500,000 courses** without slowing down. Adding a new course should not require rebuilding the whole system.

Availability (staying online)

- **99.9% uptime** — about 43 minutes of downtime per month at most.
- **If the recommendation system fails, the home page must still work** — it just shows popular courses instead of personalised ones. A user should never see a blank screen.

Compliance (following the law)

- Must follow **GDPR (Europe)** and **DPDP (India)**: users must agree before we track them, and we must delete their data within 30 days if they ask.
- Must meet **accessibility standards (WCAG 2.1 AA)** so people with disabilities can use it. We also check every 3 months that recommendations aren't unfair to any group.

Security & Data Retention

- All data **encrypted**. Instructors see only group-level numbers, never individual student names.
- **How long we keep data**: click history 13 months, learner profiles 3 years, deleted accounts wiped in 30 days, points records 2 years (in case of disputes).

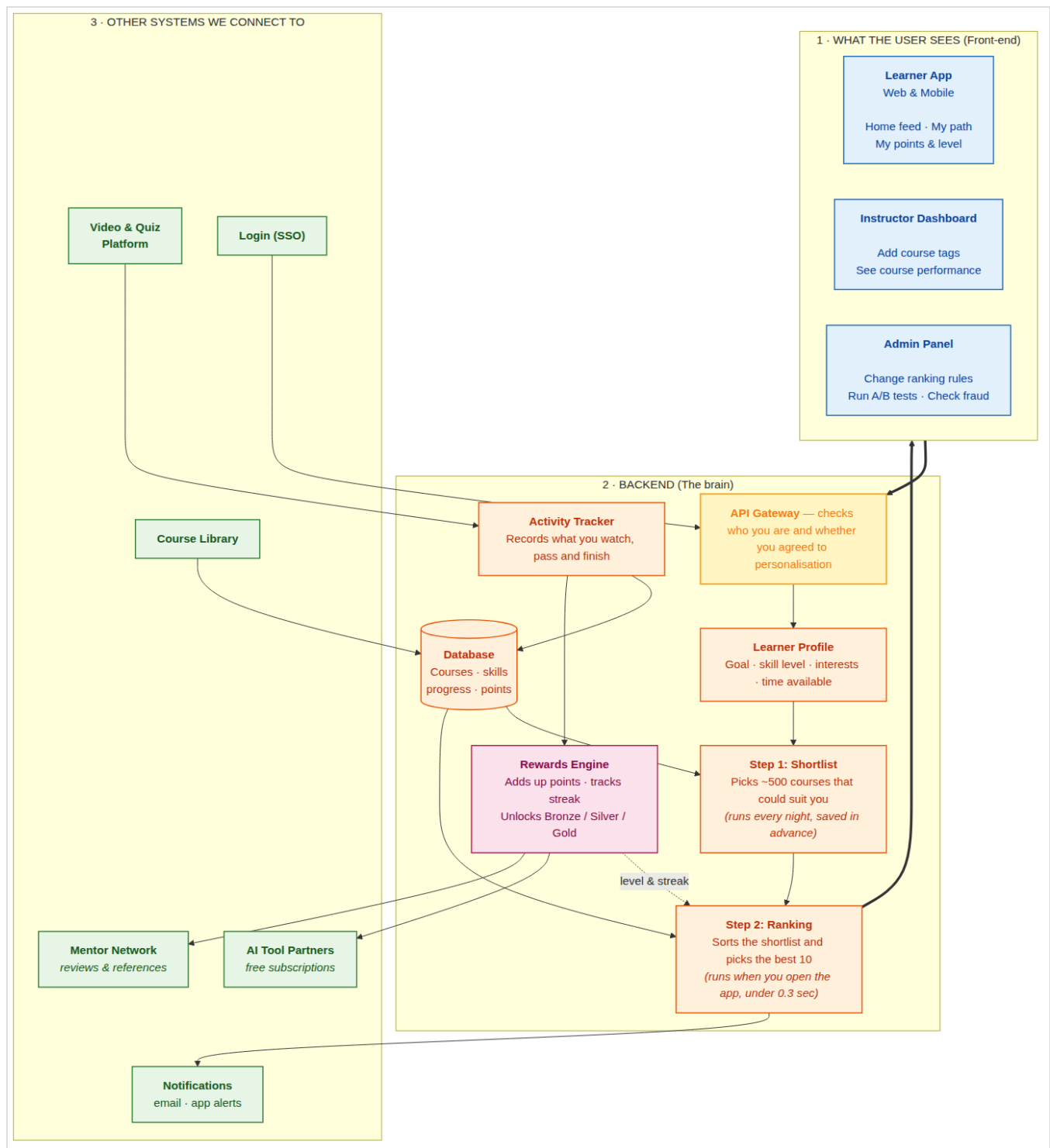
Monitoring & Logging

- **Every recommendation is logged** with the reason it was shown, so we can explain later why a learner saw a particular course.
- **Automatic alerts** if something looks wrong — for example if clicks suddenly drop 10%, or if someone is earning points suspiciously fast.

Extensibility (future-proofing)

- The recommendation model should be **swappable** — we can upgrade to a smarter model later without rebuilding the app.
 - Points rules and reward levels must be **changeable from a settings page, not from code**, because we will adjust these many times in the first year.
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Step 4 — System Design



1. User Interface — what people see

- **Learner App** (web + mobile): home feed, career path, "why this course?", points and level.
- **Instructor Dashboard**: tag courses, see performance, view the mentor review queue.
- **Admin Panel**: ranking settings, A/B tests, fairness checks, anti-cheating queue.

2. Backend — the brain

- **API Gateway** — the front door. Checks the user's login and whether they agreed to personalisation.
- **Learner Profile** — stores goal, skill level, interests and available time.
- **Step 1: Shortlist** — narrows 500,000 courses down to about 500 that could suit this learner. Runs **overnight** and is saved, because it's slow.
- **Step 2: Ranking** — sorts that shortlist and picks the **best 10**. Runs **live** when the app opens, in under 0.3 seconds.
- **Rewards Engine** — adds up points, tracks the streak, unlocks levels, and orders the AI tool subscription or mentor review.
- **Activity Tracker** — records what the learner watches, passes and finishes, and feeds it back so tomorrow's suggestions are better.
- **Database** — stores courses, skills, prerequisites, progress and points.

3. Platform Integration — systems we plug into

System	What it's for
Course Library	Where course details and tags come from
Video & Quiz Platform	Tells us what was watched, passed and completed
Login (SSO)	One account across phone and laptop, so we don't split a learner's history
Notifications	Sends reminders and "you've unlocked Silver!" messages
AI Tool Partners	Provides the free subscriptions for Silver level
Mentor Network	Manages mentors, their limits, and issuing references

How it works when Ananya opens the app

1. She opens the app → gateway checks her login and consent.
2. System loads her profile (goal: data analyst, level: beginner, 5 hrs/week).
3. Last night's shortlist of ~500 courses is pulled from storage.
4. Ranking picks the best 10 and adds a reason to each ("Because you finished SQL Basics").
5. Page loads in under 0.3 seconds.
6. She passes a quiz → Activity Tracker records it → Rewards Engine adds 10 points → her balance updates in 5 seconds → tomorrow's suggestions improve.

Step 5 — Trade-Off Analysis

The conflict: better suggestions vs. a faster app

The best possible recommendation would be calculated fresh, at the exact moment you open the app, using your entire history and a very powerful AI model.

But that would take **1-2 seconds**. And this calculation sits right on the home page — nothing loads until it finishes.

The problem: a slow app makes people leave. And "people using the app more" is exactly the goal we're trying to hit (+25% engagement). So if we chase perfect suggestions, we could actually make our own numbers *worse*.

Simply put: a fast app with good suggestions beats a slow app with perfect suggestions, because many people won't wait to see the perfect answer.

How I would prioritise

1. **Speed first.** Slowness affects 100% of users. Slightly better ranking affects only borderline cases.
2. **Spend the effort where people actually look** — the top of the page, not course number 400.
3. **Update instantly only for big events** — finishing a course, failing a quiz. Not for every scroll.

My decision

Split the work into two steps: do the slow, heavy part overnight, and only do a small, fast part when the app opens.

	What happens	When	How long
Step 1	Pick ~500 possible courses using the heavy AI model	Every night, saved in advance	Doesn't affect the user
Step 2	Sort those 500 and pick the best 10	When the app opens	Under 0.15 seconds
Backup plan	If anything fails → show popular courses instead	Automatic	Never a blank page

Why this works: most of the quality comes from picking the *right 500 courses*, which we can do overnight without the user waiting. Only the final sorting needs to be live. So we get most of the accuracy at a fraction of the speed cost — and it works well even on a slow mobile connection, which matters a lot for users in smaller cities.

What I'm giving up: if a learner watches three AI videos in one sitting, the home page won't fully rearrange itself until tonight's update. I'm accepting that, because fixing it would slow the app down for everyone, every time.

How I'd check I was right: run the "perfect live version" on 5% of users for 6 weeks. If it gets more completions *and* doesn't slow the app down, I'll change my mind.

A second trade-off: rewards vs. honest numbers

Rewards make people finish courses. But if you *pay* people to finish courses, "completion rate" stops measuring whether our recommendations are good and starts measuring how much we're paying. We could hit +30% and learn nothing.

My decision: no cash, points only for things you have to prove (quizzes, projects), a spending cap, and — importantly — **launch rewards separately from the recommendation system**, so we can tell which one caused the improvement.

What would make me switch it off: if learners who stop earning rewards then complete *fewer* courses than before, it means we replaced their real motivation with a bribe. I'd shut it down even if the headline number looked good.

Step 6 — Assumptions & Future Plans

What I'm assuming

About learners

1. Learners will answer sign-up questions if it takes under 90 seconds — but about 30% will skip it, so the system must still work for them.
2. **The main reason people quit is that the course was wrong for them**, not that the course was bad. *This is the biggest assumption in the whole project* — if people are actually quitting because of poor teaching or lack of time, better recommendations won't fix it. **I would test this first, before building anything.**
3. Learners value free AI tools and mentor access more than a small cash amount.

About our data and platform

4. We already have at least 1 year of history on what people enrolled in and finished.
5. Instructors will tag their courses properly if the tool is easy and AI suggests tags for them.
6. Our quizzes are hard enough to be worth points. If quizzes are too easy or repeat the same questions, people will cheat — so we'd need to fix quizzes *before* launching rewards.
7. The video platform sends us progress data quickly, not once a day.

About the business

8. AI tool companies will give us free subscriptions in exchange for reaching our learners. **If they won't, the Silver reward gets expensive and we'd have to redesign it.**
9. Enough senior mentors will join if we pay them a small fee and cap it at 5 references per quarter.
10. We have about 6-8 engineers, 1 data scientist, 1 designer and 1 PM for 2-3 quarters.

Launch plan

Phase	Time	What we build
Phase 0	Weeks 1–6	Set up data tracking, fix course tags, harden quizzes
Phase 1	Weeks 7–16	Sign-up questions + personalised feed → launch to 20% of users
Phase 2	Weeks 17–26	Career paths, reminders, "why this course?", instructor dashboard
Phase 3	Weeks 27–36	Admin settings, privacy controls, full launch
Phase 4	Weeks 37–48	Points, streaks and rewards

Why are rewards last? Two reasons. First, if we launch rewards and recommendations together, we'll never know which one caused the improvement. Second, paying people to push through badly matched courses is a waste — fix the matching first, then add motivation on top.

Future ideas

Soon

- Recommend a **single 15-minute lesson**, not just full courses — easier to start.
- Suggest study slots that fit the learner's actual free time.
- Spot learners who are about to quit and help them *before* they leave, instead of rewarding everyone equally.

Later

- A **chat-based advisor**: "I want to be a data analyst in 6 months, I have 5 hours a week" → it builds the plan.
- Link courses to **real job openings**, so the path leads to employment, not just a certificate.
- **Team/friend streaks** — learning with others is one of the strongest ways to finish a course.
- A **company version** where employers see team skill gaps and can co-fund the rewards.

Main risks

Risk	Why it matters	What we'd do
New learners have no history	We can't personalise for them	Use the sign-up questions to make a first guess
New courses never get shown	Instructors leave the platform	Give every new course a guaranteed trial run
Everyone gets the same popular courses	Boring, and small instructors suffer	Limit any one instructor to 40% of a feed; track variety
Bad or missing course tags	The whole system gets worse	Tagging score + AI-suggested tags before publishing
People cheat for points	Fake completions, wasted budget	Points only for proven work; daily caps; fraud alerts
Rewards get too expensive	Cost grows exactly when we succeed	Partner-funded tools, quarterly budget cap, points expire after 12 months
Too many notifications	People unsubscribe or get annoyed	One combined limit on all messages, plus quiet hours

Summary

PathFinder solves the "too many courses, no idea what to pick" problem in three moves:

1. **Ask** the learner where they want to go and test where they are now.
2. **Guide** them with a personalised, ordered path — and explain every suggestion.
3. **Motivate** them with points that unlock free AI tools, and at the very top, a real mentor's reference for their resume.

The one thing I'd defend hardest: **we keep the app fast even if that means the suggestions are slightly less perfect** — because a suggestion nobody waits to see is worth nothing.

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